

Philip Caesar Flores, PhD

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Education

Ph.D. in Physics - Supervisor: Eric A. Galapon, Ph.D.

[Quezon City](#)

UNIVERSITY OF THE PHILIPPINES DILIMAN

Aug. 2019 - Jan. 2023

- **Dissertation title:** Theory of quantized relativistic time-of-arrival operators for spin-0 particles and its application in the quantum tunneling time problem
- **Scholarships:** Department of Science and Technology-Accelerated Science and Technology Human Resource Development Program (DOST-ASTHRDP) & UP Teaching Assistantship (Teaching Fellows and Teaching Associates) Program
- **Award:** Most Outstanding PhD in Physics Graduate

M.Sc. in Physics - Supervisor: Eric A. Galapon, Ph.D.

[Quezon City](#)

UNIVERSITY OF THE PHILIPPINES DILIMAN

Aug. 2017 - Jun. 2019

- **Thesis title:** Weak equivalence principle in the quantum regime: Compatibility with quantum mechanics, and tunneling effects via the different quantizations of the time of arrival operator

B.Sc. in Physics - Supervisor: Eric A. Galapon, Ph.D.

[Quezon City](#)

UNIVERSITY OF THE PHILIPPINES DILIMAN

Jun. 2012 - Jun. 2017

- **Thesis title:** Synchronization of quantum and classical clocks, and energy translation using resolvent functional calculus for the confined time of arrival operators
- **Awards:** Leticia Shahani Award for Best Undergraduate Thesis in Physics, and Diliman BPI-DOST Science Award

Experience

Postdoctoral Researcher

[Berlin](#)

MAX BORN INSTITUTE FOR NONLINEAR OPTICS AND SHORT PULSE SPECTROSCOPY

Mar. 2023 - present

- Project: ULISSES - Ultrafast molecular chirality: twisting light to twist electrons on ultrafast time scale
- Principal Investigator: Prof. Dr. Olga Smirnova
- Develop geometric concepts in spin-resolved photoionization of chiral molecules

Teaching Fellow

[Quezon City](#)

NATIONAL INSTITUTE OF PHYSICS, UNIVERSITY OF THE PHILIPPINES DILIMAN

Sept. 2022 - Jan. 2023

- Scholarship offered by UP to PhD students in order to reduce the teaching load of the faculty members
- Responsibilities include assisting faculty members in teaching classes

Science Research Specialist

[Quezon City](#)

NATIONAL INSTITUTE OF PHYSICS, UNIVERSITY OF THE PHILIPPINES DILIMAN

Aug. 2019 - Dec. 2019

- Project: Standards and Testing Automated Modular Platform (Stamp)
- Principal Investigator: Giovanni Tapang, Ph.D.
- Develop a cheap alternative for existing ISO methodologies implemented in Regional Standards and Testing Laboratories (RSTLs).

Instructor

[Quezon City](#)

NATIONAL INSTITUTE OF PHYSICS, UNIVERSITY OF THE PHILIPPINES DILIMAN

Aug. 2017 - Jul. 2019

- Taught classes on Classical Mechanics, Optics and Electricity and Magnetism, Thermodynamics, Special Relativity, and Quantum Mechanics, and Modern Physics.
- Taught laboratory classes on Classical Mechanics, Optics Electricity and Magnetism, and Thermodynamics, Special Relativity, and Quantum Mechanics.

MISCELLANEOUS

Referee, The European Physical Journal Plus, Physical Review A, Physics Letters A, Scientific Reports, Proceedings of the Physics Society of the Philippines

Mentorship, Volunteer mentor at GradMAP Philippines: STEM Mentorship Network

Science Communication, Liknayan Podcast: Usapang Physics with Pinoy Physicists

Research

PREPRINTS

2026

20. Attosecond spin dynamics in atoms and molecules, S Carlström, [PCM Flores](#), S Patchkovskii, F Morales, O Smirnova, and M Ivanov, Roadmap on Attosecond Science, arXiv:2604.15085, p100

[arXiv](#)

2026	Geometric mechanisms enabling spin- and enantio-sensitive observables in one photon ionization of chiral molecules , PCM Flores , S Carlström, S Patchkovskii, M Ivanov, AF Ordonez, and O Smirnova, arXiv:2603.02735	arXiv
2025	Spin-current correlations in photoionization of chiral molecules , PCM Flores , S Carlström, S Patchkovskii, M Ivanov, AF Ordonez, and O Smirnova, arXiv:2505.23460	arXiv
2025	Enantiosensitive molecular compass , PCM Flores , S Carlström, S Patchkovskii, M Ivanov, V Mujica, AF Ordonez, and O Smirnova, arXiv:2505.22433	arXiv

PUBLICATIONS

2024	Instantaneous tunneling time within the theory of time-of-arrival operators , PCM Flores , DAL Pablico, and EA Galapon, Physical Review A 110 (6), 062223	PRA
2024	Partial and full tunneling processes across potential barriers , PCM Flores , DAL Pablico, and EA Galapon, Europhysics Letters 145 (6), 65002	EPL
2023	Quantized relativistic time-of-arrival operators for spin-0 particles and the quantum tunneling time problem , PCM Flores , and EA Galapon, The European Physical Journal Plus, 138, 375	EPJP
2023	Instantaneous tunneling of relativistic massive spin-0 particles , PCM Flores , and EA Galapon, Europhysics Letters, 141(1), 10001	EPL
2022	Relativistic free-motion time-of-arrival operator for massive spin-0 particles with positive energy , PCM Flores and EA Galapon, Physical Review A 99, 042113 (2022).	PRA
2019	Quantum free-fall motion and quantum violation of the weak equivalence principle , PCM Flores and EA Galapon, Physical Review A 99, 042113 (2019).	PRA
2016	Synchronizing quantum and classical clocks made of quantum particles , PCM Flores , RCF Caballar, and EA Galapon, Physical Review A 94, 032123 (2016).	PRA

CONFERENCE PROCEEDINGS

2019	Violation of the weak equivalence principle via the Born-Jordan quantized TOA operator , PCM Flores and EA Galapon, Proceedings of the Samahang Pisika ng Pilipinas
2018	Violation of the weak equivalence principle via the time of arrival operator , PCM Flores and EA Galapon, Proceedings of the Samahang Pisika ng Pilipinas
2017	Covariance property of the confined time of arrival operators , PCM Flores and EA Galapon, Proceedings of the Samahang Pisika ng Pilipinas
2016	Synchronizing quantum and classical clocks made of quantum particles up to $\hbar^2$, PCM Flores, RCF Caballar, and EA Galapon, Proceedings of the Samahang Pisika ng Pilipinas
2015	The resolvent operators of the confined time of arrival operators , PCM Flores, RCF Caballar, and EA Galapon, Proceedings of the Samahang Pisika ng Pilipinas

Conferences & Workshops

2026	Poster , Multiphoton Processes (Gordon Research Conference), Jun 7-12 2026, Stonehill College, Massachusetts, USA	USA
2026	Contributed talk , Multiphoton Processes (Gordon Research Seminar), Jun 6-7 2026, Stonehill College, Massachusetts, USA	USA
2026	Contributed talk , DPG Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section, Mar 1-6 2026, Johannes Gutenberg University Mainz, Germany	Germany
2025	Poster , Poster, Conference on Extreme Light and Chiral Molecular Systems, Sept 2 - 5 2025, Kassel, Germany	Germany
2025	Poster , 34th International Conference on Photonic, Electronic and Atomic Collisions, Jul 29 - Aug 5 2025, Sapporo, Japan	Japan
2025	Poster , 10th International Conference on Attosecond Science and Technology, Jul 6 - 11 2025, Lund, Sweden	Sweden
2025	Poster , 15th European Conference on Atoms Molecules and Photons, Jun 30 - Jul 4 2025, Congress Innsbruck, Austria	Austria
2025	Poster , Poster, 16th Femtochemistry Conference, Jun 22 - 27 2025, Savoia Hotel, Trieste, Italy	Italy
2025	Contributed talk , LIGHT-MATTER INTERACTION: Focusing on Polariton Chemistry and Physics, May 26 - 30 2025, Cultural Centre "Jules Pascin" - the Synagogue, Bulgaria	Bulgaria
2025	Contributed talk , DPG Spring Meeting of the Matter and Cosmos Section, Mar 31 - Apr 4 2025, University of Goettingen, Germany	Germany

2025	Contributed talk , DPG Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section, Mar 9-14 2025, University of Bonn, Germany	Germany
2025	Contributed talk , Time in Quantum Theory, Sept 10-13 2024, Smolenice Castle, Slovakia	Slovakia
2024	Poster Presentation , 23rd International Conference on Ultrafast Phenomena (July 15-19)	Spain
2024	Participant , COST/ZCAM School on New Computational Methods for Attosecond Molecular Processes (May 27-31)	Spain

Grants

2022, 2019, 2018	Office of the Vice President for Academic Affairs (OVPAA) Research Dissemination Grant , Awarded to outstanding faculty and REPS researchers who receive invitations or gain acceptance to present their research papers in prestigious international conferences.	Philippines
2022	Student Research Support Fund (SRSF) - Research Dissemination Grant , Support for DOST scholars accepted for oral/poster presentation in a local or international conference	Austria
2019	Office of International Linkages (OIL) Travel Grant , Support for researchers and graduate students for paper presentation at international conferences	Israel

Affiliations

2023-present	Strong-Field Theory Group , Max Born Institute for Nonlinear Optics and Short Pulse Spectroscopy	Germany
2018-present	Samahang Pisika ng Pilipinas , Professional organization of physicists and physics educators in the Philippines.	Philippines
2014-2023	Theoretical Physics Group , Members of the group perform research in mathematical physics, computational physics, statistical mechanics, quantum mechanics, nonlinear problems, gravitational physics, and relativistic astrophysics.	Philippines

References

Prof. Dr. Olga Smirnova

PROFESSOR, TECHNISCHE UNIVERSITÄT BERLIN & HEAD, STRONG FIELD THEORY GROUP, MAX BORN INSTITUTE

- smirnova@mbi-berlin.de

Prof. Dr. Misha Ivanov

PROFESSOR, HUMBOLDT UNIVERSITY & HEAD, THEORY DEPARTMENT, MAX BORN INSTITUTE

- mikhail.ivanov@mbi-berlin.de

Eric A. Galapon, PhD

PROFESSOR, UNIVERSITY OF THE PHILIPPINES DILIMAN

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